PyEXPINT: Supplementary Material

Scope. This supplement contains the development records and the complete tables behind the main text. The numerical panels are produced by `figure_sources/make_figures.py`. The script `tables/make_tables.py` regenerates the catalogue, Lawson, startup and development step-doubling tables; other tables are supplied as LaTeX with the associated records where available. Supplementary Table S6 is generated separately from the unchanged cross-check summary by `tables/make_crosscheck_table.py`. These are postprocessing operations, not new integrator executions. The development records are retained because they document the design decisions; their protocols differ from the canonical ones and they are not substitutes for the experiments of the main text.

Development crossovers

These records explain why interchangeable realizations and machine-local measurement are useful; they are not substituted for the canonical campaigns. The three panels have different measurement protocols, and the selector is excluded from these time ratios so that a hidden difference between a selected backend and an explicit backend is not presented as selector overhead. A ratio of medians is not a median of paired ratios; the canonical campaigns of the main text use the latter.

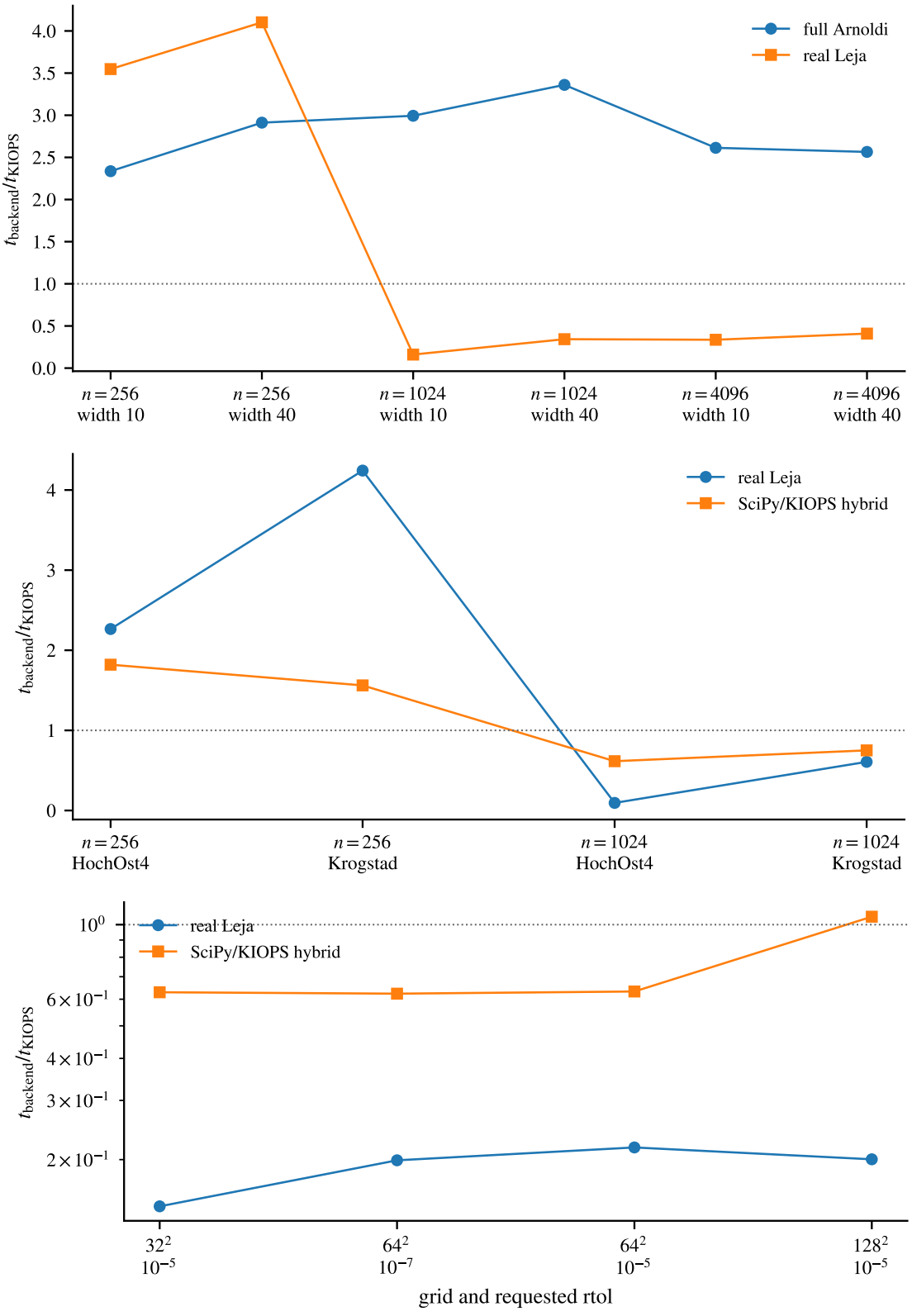


Figure S1: Development timing records not used in the canonical campaigns. Top: six one-dimensional cases with the Krogstad scheme and scaled spectral widths 10 and 40 (`crossover_phase5.json`). Middle: the hybrid-backend comparison (`hybrid_phase6.json`). Bottom: the pre-release two-dimensional experiment with three measured repetitions, recorded with Python 3.13.5, NumPy 2.3.5 and SciPy 1.17.0 rather than the production stack (`prerelease_2d_selected_fields.json`). In every panel the denominator is the KIOPS-type time of the same case. No crossover boundary is inferred between points, and these ratios are neither paired estimates nor backend rankings.

Fixed-step method campaign

Figure S2 shows the complete fixed-step campaign of the main text: 511 interior points, fixed KIOPS-type action tolerance 10^{-13}, exact history, the common numerical window $[0.5, 1]$, 27 methods and four step sizes, 108 rows in total (`method_wp_replay2341311_all.csv`). Enlarged open markers identify points at or below the precision cutoff 5×10^{-12}, tied to the action tolerance, and are excluded from the non-dominated curves.

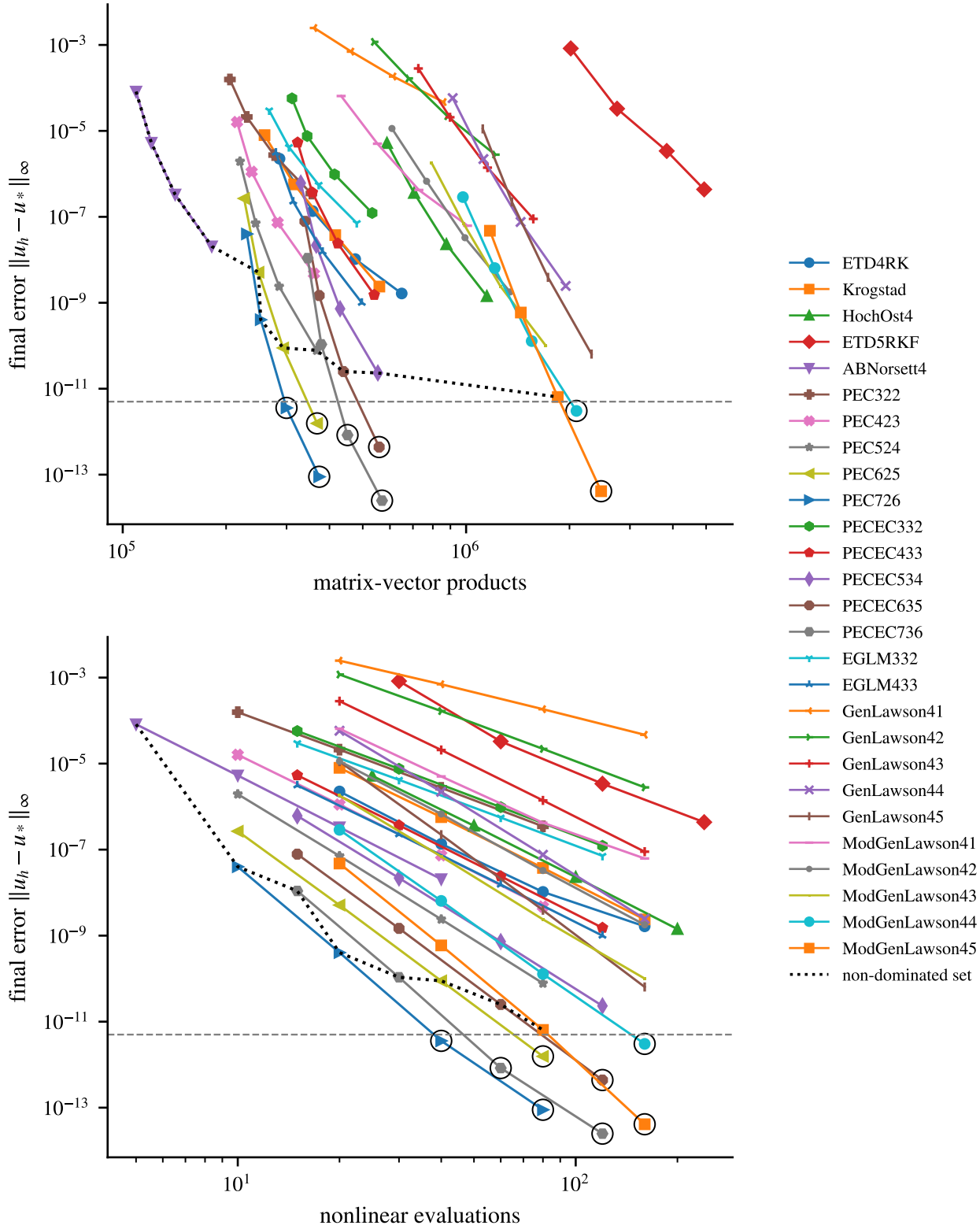


Figure S2: The fixed-step campaign with all 27 methods and four step sizes. Top: final error versus matrix-vector products. Bottom: final error versus nonlinear evaluations. The legend on the right applies to both panels; the eight methods selected in the main text are a subset of these curves.

Temporal and action tolerances

Figure S3 retains the tolerance-coupling sweep that motivated the separation of temporal and action tolerances in the adaptive solvers. It was produced with the generic step-doubling controller (`solve_adaptive`) and the Krogstad scheme on a periodic problem with 128 unknowns and $T = 1$ (`action_tolerance.json`); the action tolerance is a fraction f of the requested relative tolerance, and the achieved error and the recorded products are shown against f. ETD34 uses the same coupling policy, written explicitly here to avoid a fragile equation-number reference to the separate article:

$$\tau_{\mathrm{act}} = \min(\tau_{\max}, \max(\tau_{\min}, f\,\mathrm{rtol})).$$

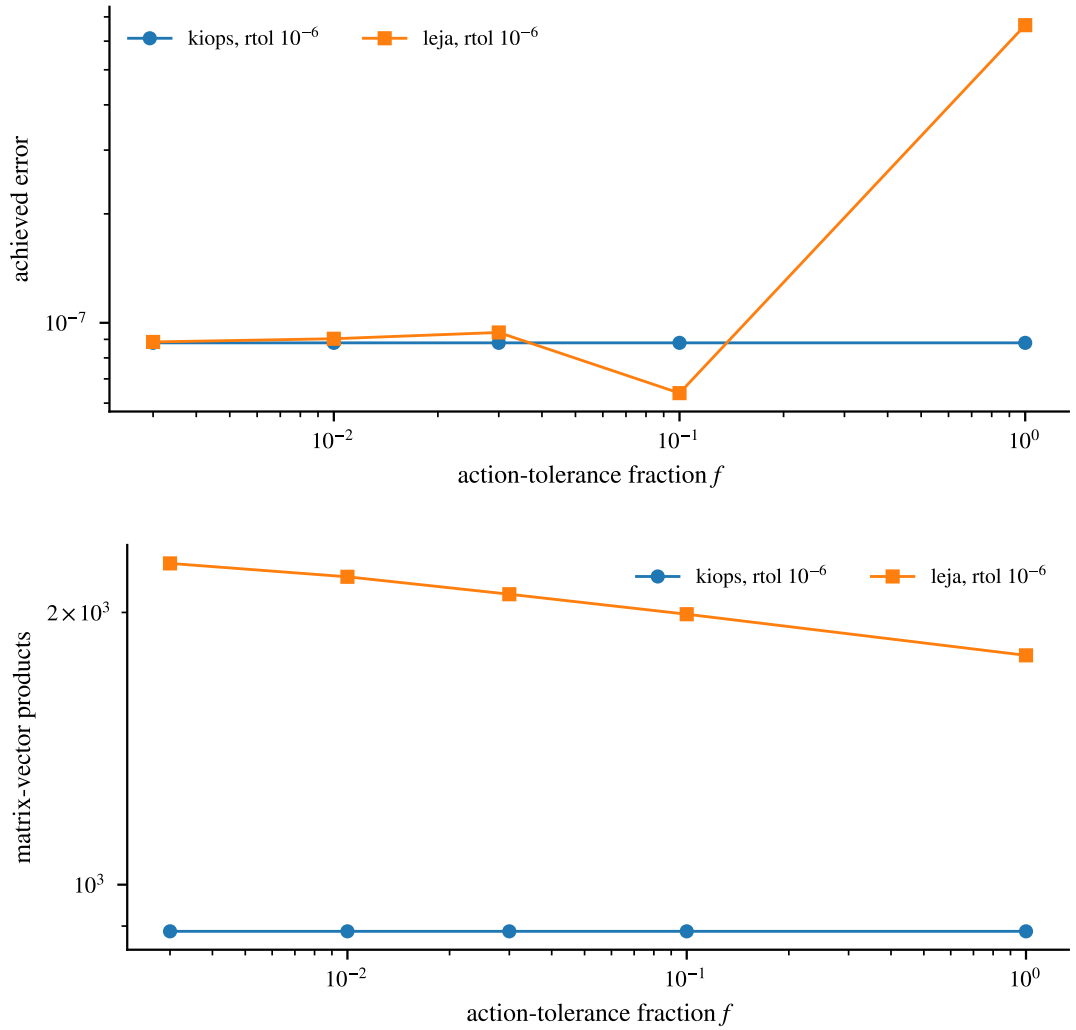


Figure S3: Tolerance-coupling sweep with the step-doubling controller. Top: achieved error. Bottom: recorded matrix-vector products. Both versus the action fraction f for the KIOPS-type and real-Leja realizations.

Catalogue coverage

Table S1 is the complete record of the 47 entries, with the provenance of their canonical coefficients, their order metadata and the outcome of the cross-check against the original EXPINT file of the same name. The complementary machine-readable index `reproducibility/coverage/catalogue_test_map.csv` lists the source definitions and concrete test identifiers supporting each entry, with the limitations described in its README.

Table S1: Metadata and recorded comparison classification of the 47 catalogue entries. p: classical order; p_s: recorded stiff order; p_w: explicitly recorded weak or conditional stiff order; s: stages; L: external layout, S solution only or H with nonlinear history; Q, P: stage and quadrature orders where the construction uses them. Prov.: provenance of the canonical coefficients, P published formula transcribed, D derived from the published construction in exact arithmetic, E taken from the EXPINT documentation. X: classification recorded in the supplied metadata, A agreement, C agreement after a one-line correction, R regenerated coefficients. The supplied summary supports the aggregate count of 42 A entries and names the C/R exceptions; it does not enumerate all 42 comparisons. This table is not a generator-to-test coverage map. A dash is an undeclared field. Generated by `tables/make_tables.py` from `catalogue_metadata.json`.

Entry	Family	p	p_s	p_w	s	L	Q	P	Prov.	X
LawsonEuler	Lawson /integrating factor Runge-Kutta	1	1	—	1	S	—	—	P	A
ABLawson2	Adams-Bashforth-Lawson	2	1	—	1	H	—	—	P	C
ABLawson3	Adams-Bashforth-Lawson	3	1	—	1	H	—	—	P	A
ABLawson4	Adams-Bashforth-Lawson	4	1	—	1	H	—	—	P	A
Lawson2a	Lawson /integrating factor Runge-Kutta	2	1	—	2	S	—	—	P	A
Lawson2b	Lawson /integrating factor Runge-Kutta	2	1	—	2	S	—	—	P	A
Lawson4	Lawson /integrating factor Runge-Kutta	4	1	—	4	S	—	—	P	A
NorsettEuler	Exponential Euler	1	1	—	1	S	—	1	P	A
ABNorsett2	Exponential Adams-Bashforth	2	2	—	1	H	—	2	P	A
ABNorsett3	Exponential Adams-Bashforth	3	3	—	1	H	—	3	P	A
ABNorsett4	Exponential Adams-Bashforth	4	4	—	1	H	—	4	P	A
PEC322	Exponential GLM /PEC	3	3	—	2	H	2	3	P	A
PEC423	Exponential GLM /PEC	4	4	—	2	H	3	4	P	A
PEC524	Exponential GLM /PEC	5	5	—	2	H	4	5	P	A
PEC625	Exponential GLM /PEC	6	6	—	2	H	5	6	P	A
PEC726	Exponential GLM /PEC	7	7	—	2	H	6	7	P	A
PECEC332	Exponential GLM / PECEC	3	3	—	3	H	2	3	P	A
PECEC433	Exponential GLM / PECEC	4	4	—	3	H	3	4	P	A
PECEC534	Exponential GLM / PECEC	5	5	—	3	H	4	5	P	A
PECEC635	Exponential GLM / PECEC	6	6	—	3	H	5	6	P	A
PECEC736	Exponential GLM / PECEC	7	7	—	3	H	6	7	P	A
EGLM332	Exponential GLM	3	3	—	3	H	2	4	E	A
EGLM433	Exponential GLM	4	4	—	3	H	3	5	E	A
ETD2RK	Exponential Runge-Kutta	2	2	—	2	S	1	2	P	C
ETD3RK	Exponential Runge-Kutta	3	3	—	3	S	—	—	P	A
EhleLawson	Exponential Runge-Kutta	2	2	—	4	S	—	—	P	A
ETD4RK	Exponential Runge-Kutta	4	2	—	4	S	1	4	P	A
Krogstad	Exponential Runge-Kutta	4	3	—	4	S	1	4	P	A
StrehmelWeiner	Exponential Runge-Kutta	4	3	—	4	S	—	—	P	A
Friedli	Exponential Runge-Kutta	4	3	—	4	S	—	—	P	A
HochOst4	Exponential Runge-Kutta	4	4	—	5	S	1	4	P	A
ETD5RKF	Exponential Runge-Kutta (nonstiff order 5)	5	—	—	6	S	—	—	E	C

Table S1 continued

Entry	Family	p	p_s	p_w	s	L	Q	P	Prov.	X
RKMK2e	Affine Lie-group	2	1	2	2	S	—	—	P	A
ETD2CF3	Affine Lie-group /ETD	3	3	—	3	S	—	—	P	C
Cfree4	Affine Lie-group / commutator-free	4	2	—	4	S	—	—	P	A
RKMK4t	Affine Lie-group /RKMK	4	1	2	4	S	—	—	P	A
GenLawson41	Generalized Lawson / exponential GLM	4	2	—	4	S	1	1	D	A
GenLawson42	Generalized Lawson / exponential GLM	4	3	—	4	H	2	2	D	A
GenLawson43	Generalized Lawson / exponential GLM	4	4	—	4	H	3	3	D	A
GenLawson44	Generalized Lawson / exponential GLM	4	4	—	4	H	4	4	D	A
GenLawson45	Generalized Lawson / exponential GLM	5	5	—	4	H	5	5	D	A
ModGenLawson41	Modified generalized Lawson /exponential GLM	4	2	—	4	S	1	2	D	A
ModGenLawson42	Modified generalized Lawson /exponential GLM	4	3	—	4	H	2	3	D	A
ModGenLawson43	Modified generalized Lawson /exponential GLM	4	4	—	4	H	3	4	D	A
ModGenLawson44	Modified generalized Lawson /exponential GLM	5	5	—	4	H	4	5	D	A
ModGenLawson45	Modified generalized Lawson /exponential GLM	6	6	—	4	H	5	6	D	R
CrankNicolson	Historical comparator /fixed-point Crank-Nicolson	2	—	—	4	S	—	—	E	A

Generic convergence of the generalized Lawson families

Table S2 reports the classical convergence of the generalized and modified generalized Lawson entries, with Krogstad, HochOst4 and ETD5RKF as controls, on a generic six-dimensional system with exact history. The slopes of GenLawson44 and GenLawson45 decrease towards their recorded orders 4 and 5 here, whereas on the parabolic problem of the main text they rise towards 5 and 6; the recorded orders are kept and the parabolic behaviour is reported as problem-dependent.

Table S2: Generic classical convergence of the generalized Lawson families and three controls. Six-dimensional system $y' = Ay + 0.3 \sin y + f(t)$ with a random symmetric negative definite A (seeds 1729 and 2718), manufactured solution $y_i = a_i \cos(\omega_i t + p_i)$, dense exact actions, exact history and window $[0.8, 1.6]$; every method integrates the window with its own steps $h = 0.2, 0.1, 0.05, 0.025$. Slopes are adjacent observed orders; the last column is the error at $h = 0.025$. Generated by `tables/make_tables.py` from `lawson_generic.json`, the output of `benchmarks/lawson_generic.py`.

Seed	Method	p	$0.2 \to 0.1$	$0.1 \to 0.05$	$0.05 \to 0.025$	Error
1729	Krogstad	4	3.91	3.96	3.98	2.70e-09
1729	HochOst4	4	3.99	4.00	4.00	2.53e-09
1729	ETD5RKF	5	4.91	4.97	4.99	2.25e-08
1729	GenLawson41	4	4.01	4.01	4.01	1.85e-08
1729	GenLawson42	4	4.28	4.16	4.08	9.05e-09
1729	GenLawson43	4	4.21	4.09	4.06	2.58e-09
1729	GenLawson44	4	4.94	4.46	4.23	1.17e-09
1729	GenLawson45	5	5.79	5.36	5.02	6.53e-11
1729	ModGenLawson41	4	4.03	4.01	4.01	5.40e-09
1729	ModGenLawson42	4	4.08	4.07	4.04	2.91e-09
1729	ModGenLawson43	4	4.38	4.25	4.15	9.99e-10
1729	ModGenLawson44	5	4.98	5.03	5.03	9.57e-11
1729	ModGenLawson45	6	6.70	6.33	6.11	2.29e-12
2718	Krogstad	4	4.00	4.00	4.00	9.39e-10
2718	HochOst4	4	4.00	4.00	4.00	1.16e-09
2718	ETD5RKF	5	4.99	5.00	5.00	2.86e-09
2718	GenLawson41	4	4.04	4.02	4.01	4.44e-09
2718	GenLawson42	4	4.20	4.10	4.05	1.48e-09
2718	GenLawson43	4	4.48	4.29	4.16	1.08e-09
2718	GenLawson44	4	5.39	4.73	4.39	7.55e-11
2718	GenLawson45	5	5.54	5.36	5.21	1.31e-11
2718	ModGenLawson41	4	4.01	4.01	4.01	1.98e-09
2718	ModGenLawson42	4	4.26	4.15	4.08	1.22e-09
2718	ModGenLawson43	4	4.51	4.40	4.26	1.28e-10
2718	ModGenLawson44	5	5.61	5.35	5.19	9.27e-12
2718	ModGenLawson45	6	5.80	6.05	6.04	2.33e-13

Canonical timing summaries

Table S3 lists the 72 configuration summaries of the three canonical campaigns, one row per replica, grid, tolerance and backend, with the achieved error and the work counters beside the time (`canonical_summary_72.csv`). The paired ratios of the main text are computed within blocks from the raw repetitions (`canonical_paired_ratios.csv`), not from these medians.

Table S3: All 72 canonical configuration summaries. Times are seconds; the displayed IQR is dispersion of solve time within a configuration, not uncertainty on the paired ratio in Figure 5. M: recorded matvecs; N: registered nonlinear evaluations.

Rep.	Side	rtol	Backend	Median	IQR	Error	M	N
1	64	1×10^{-7}	KIOPS	0.11990	0.00036	7.01×10^{-9}	542	29
1	64	1×10^{-7}	Leja	0.15031	0.00034	7.00×10^{-9}	884	29
1	64	1×10^{-7}	Hybrid†	0.12883	0.00021	7.01×10^{-9}	400	29
1	64	1×10^{-7}	Auto	0.15386	0.00053	7.00×10^{-9}	884	29
1	64	1×10^{-5}	KIOPS	0.05057	0.00015	1.46×10^{-7}	226	13
1	64	1×10^{-5}	Leja	0.06496	0.00037	1.46×10^{-7}	382	13
1	64	1×10^{-5}	Hybrid†	0.05643	0.00031	1.46×10^{-7}	170	13
1	64	1×10^{-5}	Auto	0.06679	0.00029	1.46×10^{-7}	382	13
1	128	1×10^{-7}	KIOPS	0.48228	0.00275	7.00×10^{-9}	542	29
1	128	1×10^{-7}	Leja	0.52028	0.00270	6.96×10^{-9}	1109	29
1	128	1×10^{-7}	Hybrid†	0.50336	0.00292	7.00×10^{-9}	400	29
1	128	1×10^{-7}	Auto	0.52837	0.00203	6.96×10^{-9}	1109	29
1	128	1×10^{-5}	KIOPS	0.20184	0.00130	1.46×10^{-7}	226	13
1	128	1×10^{-5}	Leja	0.22415	0.00079	1.45×10^{-7}	477	13
1	128	1×10^{-5}	Hybrid†	0.22307	0.00076	1.46×10^{-7}	170	13
1	128	1×10^{-5}	Auto	0.22820	0.00027	1.45×10^{-7}	477	13
1	256	1×10^{-7}	KIOPS	2.14766	0.01033	7.00×10^{-9}	558	29
1	256	1×10^{-7}	Leja	3.14741	0.01025	6.99×10^{-9}	1720	29
1	256	1×10^{-7}	Hybrid†	2.67566	0.01775	7.00×10^{-9}	420	29
1	256	1×10^{-7}	Auto	3.19152	0.01358	6.99×10^{-9}	1720	29
1	256	1×10^{-5}	KIOPS	0.97606	0.00554	1.46×10^{-7}	247	13
1	256	1×10^{-5}	Leja	1.41482	0.01299	1.40×10^{-7}	768	13
1	256	1×10^{-5}	Hybrid†	1.33260	0.00800	1.46×10^{-7}	180	13
1	256	1×10^{-5}	Auto	1.42320	0.00630	1.40×10^{-7}	768	13
2	64	1×10^{-7}	KIOPS	0.14570	0.00157	7.01×10^{-9}	542	29
2	64	1×10^{-7}	Leja	0.18982	0.00211	7.00×10^{-9}	884	29
2	64	1×10^{-7}	Hybrid†	0.16339	0.00329	7.01×10^{-9}	400	29
2	64	1×10^{-7}	Auto	0.19383	0.00180	7.00×10^{-9}	884	29
2	64	1×10^{-5}	KIOPS	0.06119	0.00066	1.46×10^{-7}	226	13
2	64	1×10^{-5}	Leja	0.08132	0.00103	1.46×10^{-7}	382	13
2	64	1×10^{-5}	Hybrid†	0.07148	0.00097	1.46×10^{-7}	170	13
2	64	1×10^{-5}	Auto	0.08433	0.00090	1.46×10^{-7}	382	13
2	128	1×10^{-7}	KIOPS	0.56225	0.00192	7.00×10^{-9}	542	29
2	128	1×10^{-7}	Leja	0.63236	0.00314	6.96×10^{-9}	1109	29
2	128	1×10^{-7}	Hybrid†	0.60874	0.00371	7.00×10^{-9}	400	29
2	128	1×10^{-7}	Auto	0.64149	0.00250	6.96×10^{-9}	1109	29
2	128	1×10^{-5}	KIOPS	0.23628	0.00100	1.46×10^{-7}	226	13
2	128	1×10^{-5}	Leja	0.27266	0.00304	1.45×10^{-7}	477	13
2	128	1×10^{-5}	Hybrid†	0.27236	0.00144	1.46×10^{-7}	170	13
2	128	1×10^{-5}	Auto	0.27630	0.00226	1.45×10^{-7}	477	13
2	256	1×10^{-7}	KIOPS	2.60780	0.03510	7.00×10^{-9}	558	29
2	256	1×10^{-7}	Leja	3.71156	0.02016	6.99×10^{-9}	1720	29
2	256	1×10^{-7}	Hybrid†	3.34489	0.02070	7.00×10^{-9}	420	29
2	256	1×10^{-7}	Auto	3.76064	0.01776	6.99×10^{-9}	1720	29
2	256	1×10^{-5}	KIOPS	1.17682	0.00671	1.46×10^{-7}	247	13
2	256	1×10^{-5}	Leja	1.65101	0.00798	1.40×10^{-7}	768	13
2	256	1×10^{-5}	Hybrid†	1.69073	0.01467	1.46×10^{-7}	180	13
2	256	1×10^{-5}	Auto	1.66353	0.00656	1.40×10^{-7}	768	13

Table S3 continued

Rep.	Side	rtol	Backend	Median	IQR	Error	M	N
3	64	1×10^{-7}	KIOPS	0.14690	0.00162	7.01×10^{-9}	542	29
3	64	1×10^{-7}	Leja	0.19487	0.00255	7.00×10^{-9}	884	29
3	64	1×10^{-7}	Hybrid†	0.16395	0.00148	7.01×10^{-9}	400	29
3	64	1×10^{-7}	Auto	0.19917	0.00388	7.00×10^{-9}	884	29
3	64	1×10^{-5}	KIOPS	0.06185	0.00096	1.46×10^{-7}	226	13
3	64	1×10^{-5}	Leja	0.08336	0.00095	1.46×10^{-7}	382	13
3	64	1×10^{-5}	Hybrid†	0.07207	0.00165	1.46×10^{-7}	170	13
3	64	1×10^{-5}	Auto	0.08609	0.00127	1.46×10^{-7}	382	13
3	128	1×10^{-7}	KIOPS	0.56638	0.00239	7.00×10^{-9}	542	29
3	128	1×10^{-7}	Leja	0.64107	0.00295	6.96×10^{-9}	1109	29
3	128	1×10^{-7}	Hybrid†	0.61430	0.00481	7.00×10^{-9}	400	29
3	128	1×10^{-7}	Auto	0.65079	0.00413	6.96×10^{-9}	1109	29
3	128	1×10^{-5}	KIOPS	0.23842	0.00256	1.46×10^{-7}	226	13
3	128	1×10^{-5}	Leja	0.27628	0.00184	1.45×10^{-7}	477	13
3	128	1×10^{-5}	Hybrid†	0.27449	0.00214	1.46×10^{-7}	170	13
3	128	1×10^{-5}	Auto	0.28100	0.00181	1.45×10^{-7}	477	13
3	256	1×10^{-7}	KIOPS	2.61837	0.02821	7.00×10^{-9}	558	29
3	256	1×10^{-7}	Leja	3.74869	0.01589	6.99×10^{-9}	1720	29
3	256	1×10^{-7}	Hybrid†	3.34672	0.01413	7.00×10^{-9}	420	29
3	256	1×10^{-7}	Auto	3.79864	0.02051	6.99×10^{-9}	1720	29
3	256	1×10^{-5}	KIOPS	1.17445	0.01325	1.46×10^{-7}	247	13
3	256	1×10^{-5}	Leja	1.68432	0.01042	1.40×10^{-7}	768	13
3	256	1×10^{-5}	Hybrid†	1.68878	0.01523	1.46×10^{-7}	180	13
3	256	1×10^{-5}	Auto	1.69008	0.00717	1.40×10^{-7}	768	13

† Hybrid counts refer to the KIOPS component and omit internal SciPy work. The reported Auto endpoint selection is Leja in these summaries. Detailed Leja degrees, all internal SciPy calls and separate backend-failure counts are not filled from later software: absent historical fields remain absent. The included normalized CSV retains numerical precision beyond this display table.

Adaptive and cost-aware diagnostics

Table S4 reports the stiff adaptive comparisons of ETD34 with step doubling on the local and nonlocal manufactured problems, and the cost-aware stress test (`tableS04a_stiff.csv`, `tableS04b_cost.csv`). As discussed in the main text, a state-independent additive source cancels directly from the same-time nonlinear difference in the ETD34 estimator. When $N(t, y) = g(t) + F(y)$, the source can still influence that difference through the stage states and F. These records do not isolate that indirect effect or establish a causal decomposition of the error.

Table S4: Additional adaptive evidence, kept within its original protocol. (a) HO problems with $n = 100$, variant 1 local and variant 2 nonlocal. (b) KIOPS-type cost-control stress test, $n = 512, T = 2$. A/R denotes recorded accepted/rejected steps. Separate backend-failure counts were not recorded in these files.

(a) Stiff adaptive comparison

Variant	rtol	Controller	Error	A/R	N calls
1	1×10^{-4}	ETD34	3.78×10^{-5}	5/0	21
1	1×10^{-4}	doubling	2.73×10^{-6}	5/0	60
1	1×10^{-6}	ETD34	7.39×10^{-7}	13/1	57
1	1×10^{-6}	doubling	1.66×10^{-6}	6/0	72
2	1×10^{-4}	ETD34	4.44×10^{-6}	5/0	21
2	1×10^{-4}	doubling	2.09×10^{-7}	5/0	60
2	1×10^{-6}	ETD34	3.51×10^{-8}	22/2	97
2	1×10^{-6}	doubling	2.67×10^{-7}	7/0	84

(b) Cost-aware control: development timings

rtol	Control	Error	Matvecs	N calls	A/R	Seconds
1×10^{-4}	accuracy	4.74×10^{-6}	3347	17	4/0	2.0438
1×10^{-4}	cost	3.69×10^{-6}	3079	33	8/0	2.2428
1×10^{-6}	accuracy	4.29×10^{-8}	6469	57	12/2	4.5922
1×10^{-6}	cost	2.56×10^{-8}	5083	105	24/2	4.7677

The cost controller reduced matvecs by about 8.0% and 21.4%, while recorded wall time increased by about 9.7% and 3.8%. These unreplicated development measurements are a diagnostic counterexample to an incomplete cost proxy, not a canonical timing comparison or a universal efficiency claim.

Step-doubling records of a development experiment

Table S5 lists the records of an earlier development experiment with the generic step-doubling controller, two requested tolerances per method and backend (`phase6_work_precision_selected_fields.json`). They are shown as a table because two points per series do not support a curve.

Table S5: Records of an earlier development experiment with the generic step-doubling controller (three complete method steps per trial, hence $3s$ nonlinear evaluations per attempt) on 128-dimensional periodic and Dirichlet problems with Krogstad and HochOst4, requested tolerances 10^{-4} and 10^{-7} (absolute tolerance 10^{-3} times rtol), and the KIOPS-type, real-Leja and calibrated selector realizations. Each row is one solve; no curve is fitted through the two tolerances. These are not ETD34 records. The calibrated selector of these runs kept its thresholds but, in the version used then, lost its calibration record when cloned by the solver, a defect corrected in the submitted version. Generated by `tables/make_tables.py`.

Problem	Method	Backend	rtol	Error	Matvecs	N evals	Acc./rej.
dirichlet	HochOst4	auto-calibrated	1e-07	1.48e-07	4299	105	6/1
dirichlet	HochOst4	auto-calibrated	1e-04	2.05e-06	1836	45	3/0
dirichlet	HochOst4	kiops	1e-07	1.48e-07	4129	105	6/1
dirichlet	HochOst4	kiops	1e-04	2.05e-06	1556	45	3/0
dirichlet	HochOst4	leja	1e-07	1.48e-07	3901	105	6/1
dirichlet	HochOst4	leja	1e-04	2.15e-06	1580	45	3/0
dirichlet	Krogstad	auto-calibrated	1e-07	1.52e-07	2031	84	6/1
dirichlet	Krogstad	auto-calibrated	1e-04	2.03e-06	864	36	3/0
dirichlet	Krogstad	kiops	1e-07	1.52e-07	1862	84	6/1
dirichlet	Krogstad	kiops	1e-04	2.03e-06	746	36	3/0
dirichlet	Krogstad	leja	1e-07	1.52e-07	2910	84	6/1
dirichlet	Krogstad	leja	1e-04	2.13e-06	1173	36	3/0
periodic	HochOst4	auto-calibrated	1e-07	2.89e-08	3452	90	5/1
periodic	HochOst4	auto-calibrated	1e-04	2.20e-07	1726	45	3/0
periodic	HochOst4	kiops	1e-07	2.89e-08	2880	90	5/1
periodic	HochOst4	kiops	1e-04	2.20e-07	1440	45	3/0
periodic	HochOst4	leja	1e-07	2.90e-08	4499	90	5/1
periodic	HochOst4	leja	1e-04	3.24e-07	1898	45	3/0
periodic	Krogstad	auto-calibrated	1e-07	2.24e-08	1552	72	5/1
periodic	Krogstad	auto-calibrated	1e-04	1.29e-07	776	36	3/0
periodic	Krogstad	kiops	1e-07	2.24e-08	1296	72	5/1
periodic	Krogstad	kiops	1e-04	1.29e-07	648	36	3/0
periodic	Krogstad	leja	1e-07	2.24e-08	3332	72	5/1
periodic	Krogstad	leja	1e-04	1.74e-07	1452	36	3/0

Archived cross-check

The unchanged report at `reproducibility/expint_crosscheck/crosscheck_report.txt` supplies Table S6. The same directory additionally contains 102 numerical-output rows, coefficient values, refinement output, the correction diffs, selected original-source hashes and copies of both comparison drivers. The drivers' original runnable location is in the private software companion, where the Python comparison resolves its source imports. Original EXPINT MATLAB files are not redistributed. The coefficient indices in the drivers are one-based. Equality at two configurations is not an order theorem.

Table S6: Numerical statements in the supplied cross-check summary. Top: relative final-state differences on the scalar and matrix problems, before and after the four reported one-line corrections. The first row gives the reported aggregate maxima for 42 entries, not individually reconstructed results. Dashes denote comparisons not supplied or not applicable, not zero. Bottom: the six coefficient pairs reported for ModGenLawson45 at $z = -0.3$; decimal values are reproduced verbatim, and the companion drivers use one-based indices.

	Original 2005 file		One-line correction	
Entry	Scalar	Matrix	Scalar	Matrix
42 entries (reported maxima)	1.3×10^{-15}	5.9×10^{-15}	—	—
ABLawson2	3.6×10^{-2}	3.7×10^{-2}	2.3×10^{-16}	6.2×10^{-15}
ETD2RK	2.5×10^{-1}	2.8×10^{-1}	0.0	5.0×10^{-15}
ETD5RKF	6.5×10^{-3}	8.5×10^{-15}	2.3×10^{-14}	8.5×10^{-15}
ETD2CF3	3.4×10^{-7}	5.3×10^{-7}	2.7×10^{-15}	1.7×10^{-15}
ModGenLawson45	7.7×10^{-9}	2.5×10^{-8}	—	—

Coefficient	PyEXPINT	EXPINT 2005
`V[1,2]`	`-0.013806254694738`	`-0.008445356126745`
`V[1,3]`	`+0.011344865515137`	`+0.005983966947144`
`V[1,4]`	`-0.004742475054817`	`-0.002062025770820`
`V[1,5]`	`+0.000806499599604`	`+0.000270409742804`
`B[1,1]`	`+0.132016135298815`	`+0.129335686014818`
`B[1,4]`	`+0.164515176113568`	`+0.165051265970368`

The summary reports 47 compared entries, 42 agreeing within 10^{-12} on both problems, four agreeing after a one-line correction, and one regenerated entry. Its final status is `RESULT: OK`. This summary table is complemented by the separately archived refinement outputs, problem definitions, correction diffs and execution-environment record. The regenerated entry has no separate corrected final-state result in the report. This table is a transcription, not a new Octave or PyEXPINT execution.

Source boundary

Selected plotting records are under `reproducibility/t8_data`; the original Allen–Cahn record in that directory is byte-identical to the workstation JSON in the private companion. Its old provisional transcription is retained only in the revision history and is not plotted. The private archive `pyexpint_review_package_20261011.zip` contains the source snapshot, wheel/sdist, tests, examples, drivers and detailed logs. The associated evidence index and coverage map identify those paths and separate original experiment outputs, logged candidate comparisons and manuscript postprocessing. No new integration, test-suite run or public software release is implied by this manuscript revision.